

Maria is a 52-year-old woman who visits her general practitioner after noticing a persistent lump in her right breast. She is referred to the hospital for further evaluation.

During her first **clinical visit**, imaging procedures (mammography and ultrasound) are performed. The radiologist identifies a suspicious lesion, and a biopsy is scheduled. The biopsy confirms **invasive ductal carcinoma**. The pathology report documents tumor grade 2 and positive hormone receptor status (ER+, PR+, HER2–).

At this point, a **Cancer Condition** is established for Maria. The condition is defined by:

- Body site: Right breast
- Histology behavior: Invasive carcinoma
- Tumor grade: Grade 2
- Biomarker profile: ER+, PR+, HER2–
- Base of assertion: Biopsy
- Asserted date: Date of pathology confirmation

Further staging investigations are performed during subsequent visits. Imaging studies show no distant metastases. The disease is classified as **localized**. This information is recorded as a state of the disease “over time and space,” capturing both the extent (local) and the date on which this assessment was made. Each assertion is linked to its evidence source (imaging reports) and tied to a specific visit.

Maria’s case is discussed in a multidisciplinary tumor board. A **treatment plan** is created with curative intent. The plan includes:

1. Lumpectomy (surgery)
2. Adjuvant radiotherapy
3. Long-term endocrine therapy

These interventions are represented as components of a single Treatment targeting Maria’s cancer condition. Each component has associated start and end dates, and the overall intent is explicitly documented as curative.

After surgery, pathology results refine the pathological stage. The cancer condition is updated based on this new evidence, again recording the base of assertion (surgical pathology report) and the date of the update. Radiotherapy follows, and Maria begins endocrine therapy.

During follow-up visits, imaging and laboratory assessments are performed. Six months after completing radiotherapy, imaging shows no evidence of residual disease. A **Treatment Response** is recorded during that visit. The response is categorized as “complete remission,” based on imaging findings. The response is explicitly linked to:

- The treatment being evaluated
- The visit during which evaluation occurred
- The evidence source (imaging report)
- The evaluation date

Over the next three years, Maria attends routine follow-up visits. Each visit includes clinical examination and imaging. Her disease status remains stable, and her life status is recorded as alive without evidence of disease.

Four years later, Maria develops persistent abdominal discomfort. During a new visit, imaging reveals lesions in the liver. A biopsy confirms metastatic breast cancer. A new disease state is recorded:

- Extent: Metastatic
- Site: Liver
- Disease status: Progression
- Base of assertion: Imaging and biopsy
- Asserted date: Date of confirmation

This does not replace her original cancer condition; rather, it represents an evolution of the disease over time and space.

A new treatment plan is initiated, this time with palliative intent. Maria starts systemic chemotherapy. As before, each treatment cycle, response assessment, and disease status update is recorded during visits and supported by documented evidence sources.

Throughout Maria’s journey, the model ensures that:

- Every diagnosis is traceable to its evidence.
- Every disease state is time-stamped and provenance-aware.

- Every treatment has a defined intent and target.
- Every treatment response is explicitly evaluated and documented.
- Disease evolution is captured longitudinally rather than overwritten.

In this way, Maria's cancer history is not just a series of disconnected records, but a coherent, temporally structured, evidence-linked narrative that supports clinical care, research analysis, and decision support.